Asymmetric Density Induced Spatial Pressure Gradient Propulsion

—— A Formal Protocol Based on "Universal Phase Cycles" and Macro-Verification via 'Oumuamua Case Study

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Abstract

This paper challenges the conventional matter-centric paradigms used to explain non-gravitational acceleration in celestial mechanics. By postulating that space-time is a physical manifold with intrinsic background pressure (Dark Energy), we derive a dynamic correlation between **asymmetric density distribution** and **spatial phase gradients**. We provide a logical audit of the interstellar object **'Oumuamua**, suggesting its anomalous trajectory originates from its geometric-density asymmetry rather than outgassing. Finally, we propose the **"Long-Bar Experiment"**—a low-cost, high-precision orbital test utilizing a kilometer-scale non-symmetric tether to provide definitive evidence of space-pressure-induced propulsion.

1. The Logical "Dead End" of Contemporary Physics

Modern astrophysics relies on "outgassing" or "radiation pressure" to explain anomalous accelerations. However, these models often fail the test of **Occam's Razor**, requiring invisible mechanisms (e.g., hydrogen/nitrogen ice sublimation) to satisfy momentum conservation. We argue that the root of the problem lies in the misinterpretation of space as a void, rather than a pressurized medium.

2. Core Theory: Universal Phase Cycle & Density Gradients

We define matter density not as a static quantity, but as a **condensed phase of spatial fibers**.

- **Spatial Background Pressure (P_{bg}):** Space-time exerts a uniform compressive pressure.
- **Density-Phase Coupling:** High-density matter displaces spatial fibers more intensely, creating a localized low-pressure phase.
- **Asymmetric Thrust Mechanism:** A structural gradient between high-density (ρ_{high}) and low-density (ρ_{low}) components induces a net directional force:

$$F_{thrust} = \oint (\vec{P}_{bg} \cdot \eta(\rho)) dA$$

where η is the phase coupling coefficient.

3. Case Audit: 'Oumuamua as a Natural Prototype

'Oumuamua's extreme 10:1 aspect ratio and smooth, non-thermal acceleration suggest it is a natural **asymmetric phase body**.

- **Hypothesis:** Internal phase separation led to a density gradient across its longitudinal axis.
- **Inference:** Its acceleration is a passive "surfing" effect on the solar system's dark energy density gradient. It consumes no fuel; it utilizes the structure of space itself.

4. Proposed Verification: The Long-Bar Experiment

To break the cycle of theoretical speculation, we propose a macro-scale verification:

1. **Structure:** A lightweight carbon-fiber tether ($\geq 1000m$).
2. **Payloads: * Head:** High-density Tungsten alloy (High-Pressure Phase).
   - **Tail:** Ultra-low-density Aerogel (Balanced Phase).
3. **Observation:** Deploying the tether in a gravitationally flat region (e.g., Lagrange Point L2).
4. **Prediced Outcome:** A persistent, non-inertial intrinsic acceleration ($a > 0$) toward the high-density end, measurable via laser interferometry.

5. Conclusion: A Prospect for Future Civilizations

If $a > 0$ is confirmed, it marks the end of the propellant-based era. Humanity will transition from "fighting" gravity to "aligning" with the respiratory cycles of the universe.